

Centennial Jing-Zhang: From Engineering Autonomy to Urban Adoption

Culture becomes an urban-adoption institution for truth, handoff, access and public learning

THE CENTENNIAL FIRST MILE | FROM ENGINEERING AUTONOMY TO URBAN AI ADOPTION



RAIL IS NOT DECORATION: LINE, STATION, MILEPOST, SIGNAL AND MAINTENANCE BECOME PUBLIC ADOPTION DISCIPLINE

1909 → 2026 → NEXT

A CENTURY AGO, THE JING-ZHANG RAILWAY
MARKED A FIRST MILE OF ENGINEERING AUTONOMY.
TODAY, JING-ZHANG FIRST MILE TAKES AI
FROM MODEL TO REAL CITY ADOPTION.



FIRST-MILE GATE

The five sleepers are operational: every gate needs evidence and allows pause, return and exit.

THE JING-ZHANG +1: turn engineering autonomy into adoption discipline for public AI services.

FIVE RAILWAY GRAMMARS, FIVE URBAN MECHANISMS

LINE	LINE	From isolated showcases to one handoff chain
STATION	STATION	Each station completes one adoption task
MILEPOST	MILEPOST	The passport records how far adoption travelled
SIGNAL	SIGNAL	G0-G4 publish proceed, pause and exit
MAINTAIN	MAINTENANCE	It becomes city service only after unfamiliar handoff

PROVISIONAL BOUNDARY · CONCEPT DESIGN · RECALCULATE AFTER OFFICIAL POLYGONS

Xiaokang × Codex Urban Agent · 2026-08

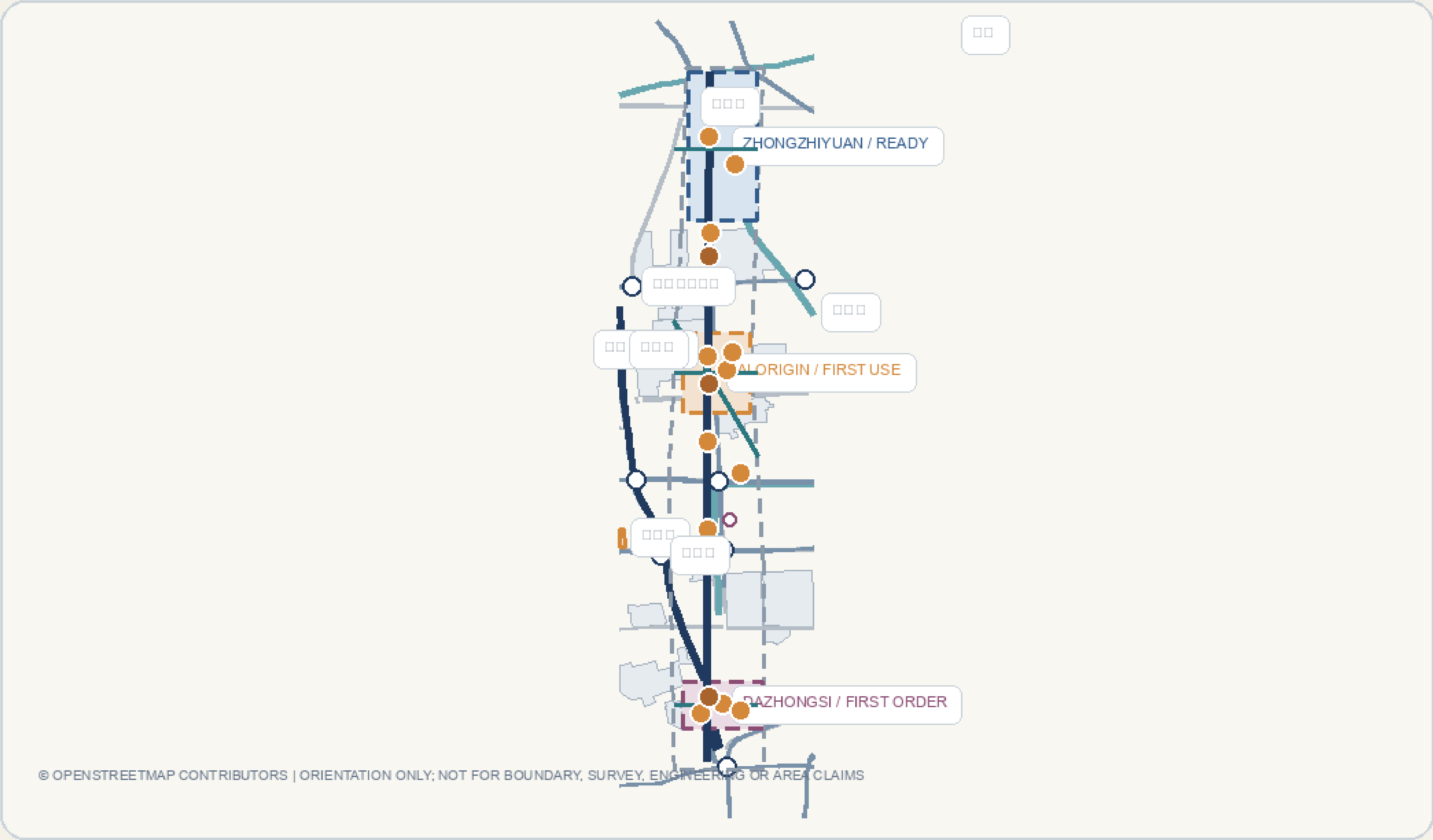
Real-Place Atlas

Ground the adoption chain in roads, rivers, rail, campuses, stations and heritage without promoting provisional extents

REAL-PLACE ATLAS | AN ADOPTION CHAIN ANCHORED IN ROADS, RIVERS, RAIL, CAMPUSES AND HERITAGE



PRESENTATION-ONLY OSM CONTEXT + PROVISIONAL DESIGN BOUNDARY; NOT SURVEY OR STATUTORY CONTROL



SIX REAL-PLACE JUDGMENTS

- 01 The heritage park is an existing everyday spine, not a future green-axis invention
- 02 The north meets Qinghe and tech campuses: fit for readiness and safe handoff
- 03 The middle meets Wudaokou, Qinghuayuan heritage and transit: fit for first users
- 04 The south meets Dazhongsi heritage, transit and commerce: fit for first order and reuse
- 05 Cross streets, campuses and communities define launch ports, not just a north-south promenade
- 06 Every AI service is removable; routes, shade, seats and human entry remain

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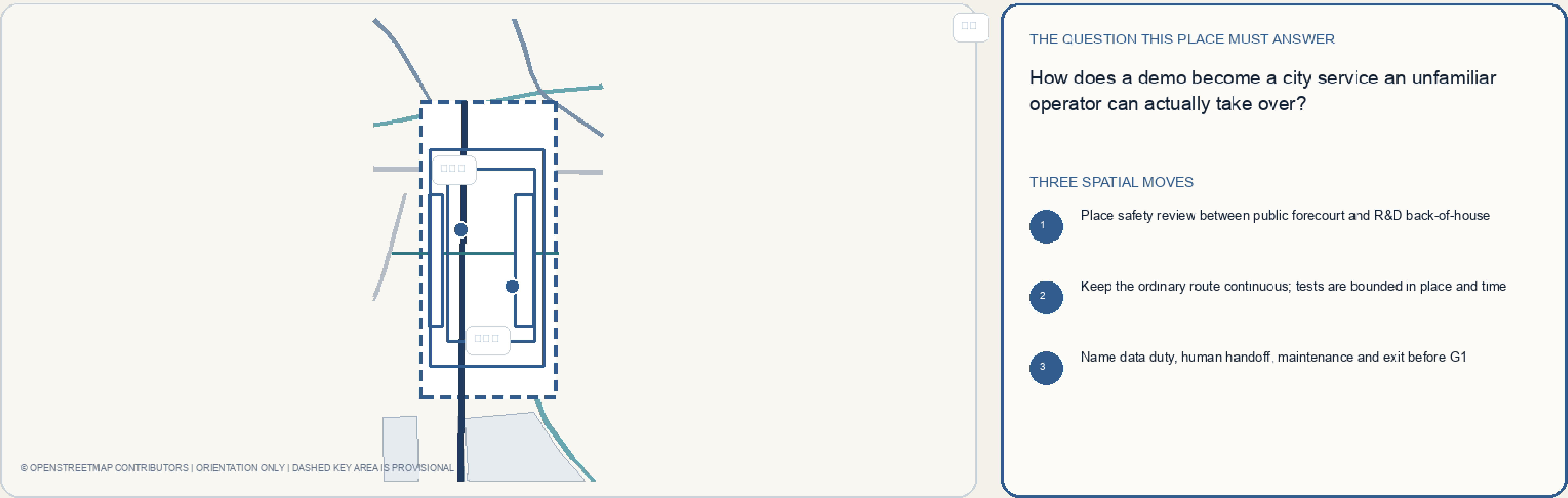
Zhongzhiyuan: Turn a Prototype into a Safely Handoff-Ready Product

A product-ready yard between Qinghe and campus clusters; no data owner, human takeover or maintenance plan means no G1

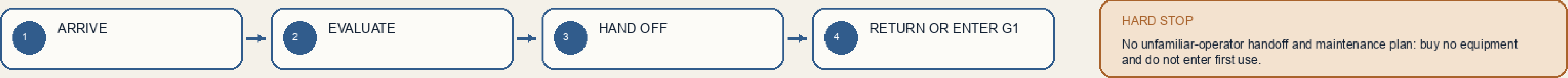
ZHONGZHIYUAN | PRODUCT READINESS AS A HANDOFF-READY URBAN FORECOURT



QINGHE-FIFTH RING-XUEQING ROAD TECH/CAMPUS EDGE | REAL ORIENTATION + PROVISIONAL KEY AREA



FIRST-MILE ACTION SEQUENCE



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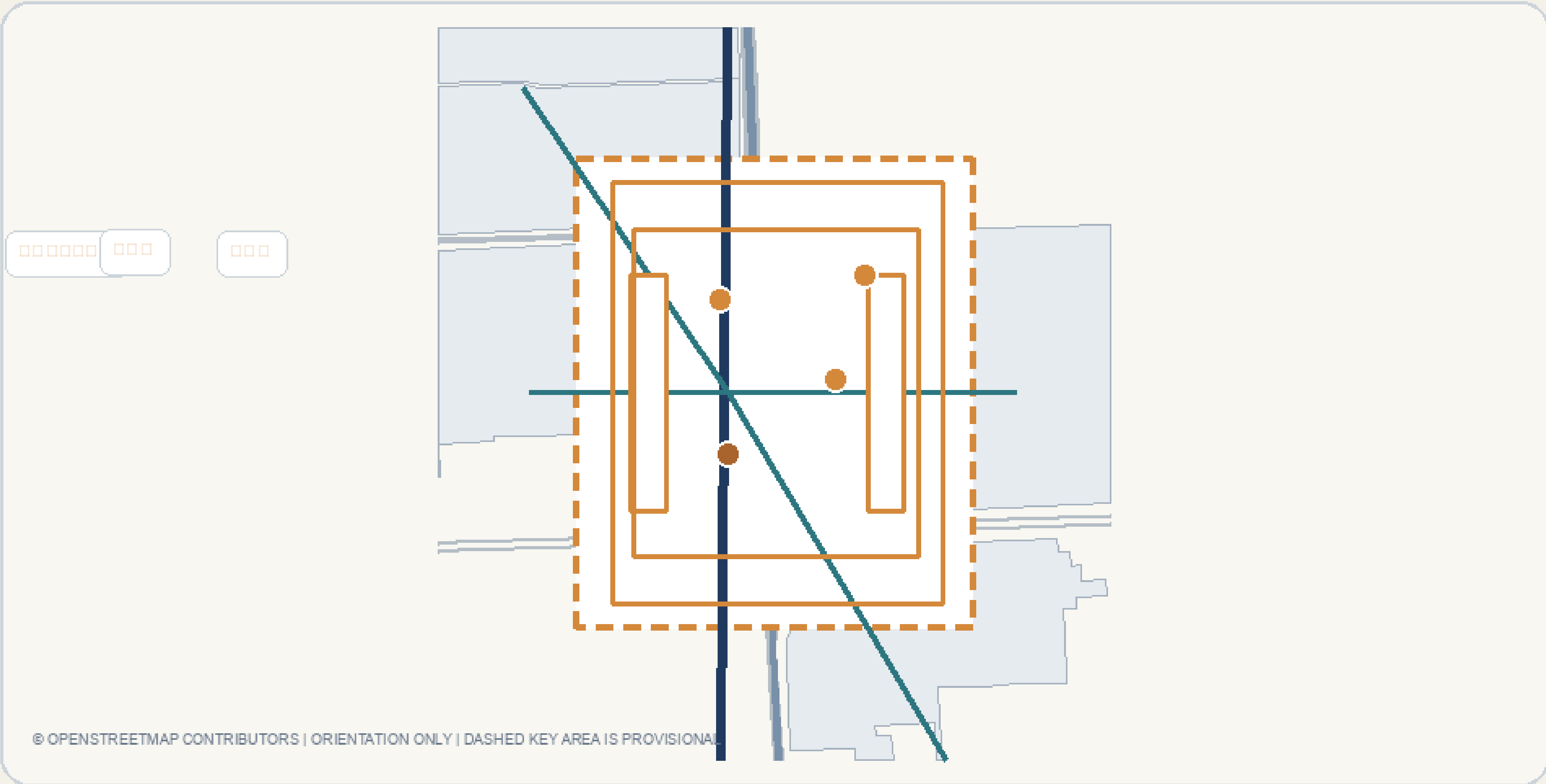
AI Origin: Turn the Everyday Route into a First-User Commons

Between Wudaokou, Qinghuayuan heritage and transit; ordinary route first, human service visible, AI optional

AI ORIGIN | TURN THE EVERYDAY ROUTE BETWEEN WUDAOKOU AND QINGHUAYUAN HERITAGE INTO A FIRST-USE COMMONS

FIRST MILE
JING-ZHANG

WUDAOKOU–CHENGFU ROAD–QINGHUAYUAN HERITAGE–LINE 13 | REAL ORIENTATION + PROVISIONAL KEY AREA



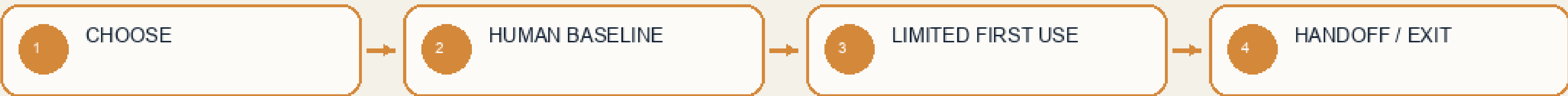
THE QUESTION THIS PLACE MUST ANSWER

Can an ordinary person complete a real task and transfer to a human without loss when AI fails?

THREE SPATIAL MOVES

- 1 Keep the 6 m ordinary route primary; AI is optional
- 2 Make the human/appeal desk visible and directly reachable
- 3 Use the First User Table to connect founders, residents and front-line staff

FIRST-MILE ACTION SEQUENCE



HARD STOP

If any of three real tasks cannot hand off or exit without loss, pause G2 immediately.

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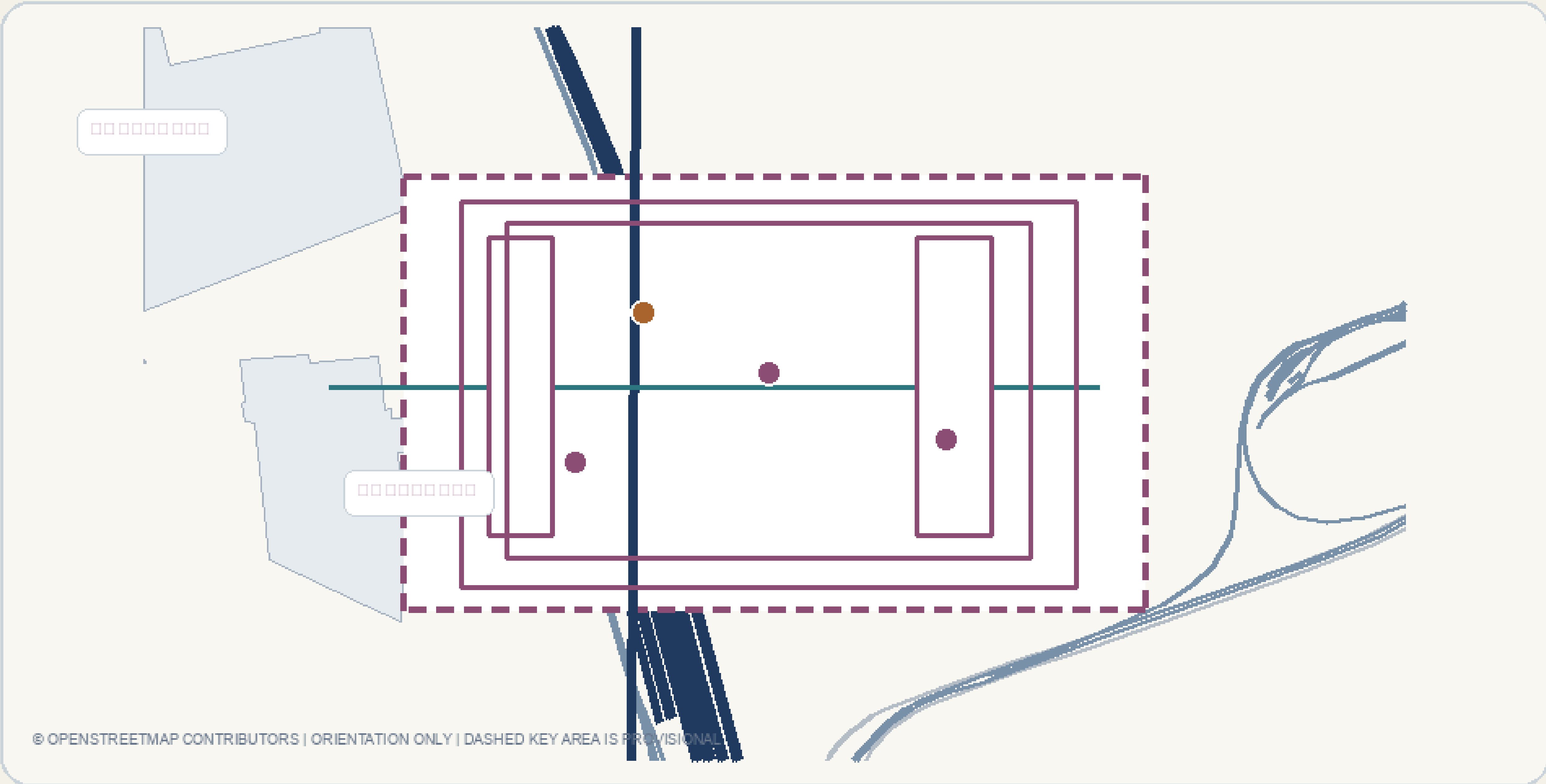
Dazhongsi: Turn Pilot Evidence into Reusable First-Order Terms

Station, heritage, commons and everyday commerce connect; a first order states acceptance, fee, liability, training and exit together

DAZHONGSI | LINK TRANSIT ARRIVAL, HERITAGE AND EVERYDAY COMMERCE INTO A VERIFIABLE FIRST-ORDER ROOM

 FIRST MILE
JING-ZHANG

DAZHONGSI STATION–NORTH THIRD RING–HERITAGE–BEIJING JIAOTONG UNIVERSITY | REAL ORIENTATION + PROVISIONAL KEY AREA



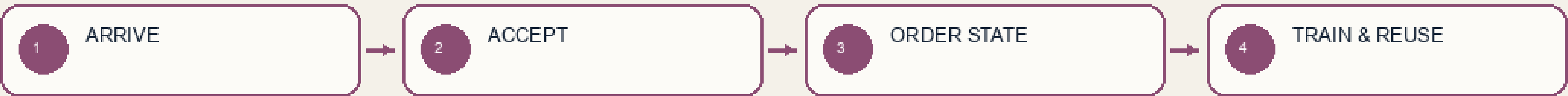
THE QUESTION THIS PLACE MUST ANSWER

How does one pilot become reusable terms with acceptance, fee, liability, training and exit?

THREE SPATIAL MOVES

- 1 Link station, public room and daily commerce in one arrival loop
- 2 Let procurement clinic and ordinary public stay coexist, not become a showroom
- 3 Address reuse/failure records to the next operator, not media traffic

FIRST-MILE ACTION SEQUENCE



HARD STOP

If acceptance, fee, liability, procurement state, training or exit is blank, do not call it a first order.

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A First User Is Not an Event Check-In

Human-scale flagship experiences are followed by a six-step, four-lane journey proving correction, human handoff and no-loss exit



FIRST-USER JOURNEY | TURN USE INTO REVIEWABLE ADOPTION EVIDENCE

SIX STEPS, FOUR SWIMLANES | PAUSE IF HUMAN ESCALATION OR NO-LOSS EXIT FAILS

	01 ARRIVE & CHOOSE	02 DEFINE REAL TASK	03 HUMAN BASELINE	04 LIMITED AI USE	05 CORRECT & HANDOFF	06 EXIT, REUSE OR STOP
FIRST USER	See 6m ordinary route; choose human or AI	State today's decision; 'experience' is not a task	Run current process once; time errors and material	Rights-cleared public info only; stop anytime	Flag error; transfer without losing progress	Receive recoverable record; end or reuse
FRONT-LINE STAFF	Explain boundary and non-digital entry	Confirm owner, done state and harm	Record human steps and waiting	Stay visible; may overturn the answer	Take over, correct and close the loop	Confirm no-loss service; explain appeal/deletion
AI ASSISTANCE	Off by default; no face/trajectory	Help decompose, never decide	Read de-identified baseline only	Show source, confidence, refusal and handoff	Export context; stop automation	No departure tracking; delete on schedule
EVIDENCE & GATE	G0: choice visible	Task card: owner / done / harm	Baseline: time / error / handoff	G2: source / refusal / consent	G3: handoff / correction / complaint	G4: outcome / unit cost / reuse / exit

SUCCESS IS NOT SATISFACTION: RECORD COMPLETION, HUMAN CORRECTION, INDEPENDENT HANDOFF, UNIT COST, NO-LOSS EXIT AND CONDITIONAL REUSE.

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The AI-generated triptych is concept communication, not site, interview, partnership, order or built evidence; status follows the Passport and G0-G4 records.

Twelve Scenario Cards: Every One Must Pass the Gates

The chain starts only with a real task, human baseline, minimal data, operating owner, cost state and stop condition

TWELVE SCENARIO CARDS | EVERY ONE MUST PASS THE FIRST-MILE GATES



REAL TASK, HUMAN BASELINE, MINIMAL DATA, OPERATING OWNER, COST STATE, HANDOFF AND STOP CONDITION ARE ALL REQUIRED

01

Urban Problem Launch Desk

Real-need intake | Human moderation

HUMAN HANDOFF / PAUSE / EXIT

02

Pre-Street Model Evaluation

Fact, bias and safety | Expert final review

HUMAN HANDOFF / PAUSE / EXIT

03

Agent Operations Handoff Pod

Unfamiliar operator takeover | Return on failure

HUMAN HANDOFF / PAUSE / EXIT

04

AI Startup Service Copilot

Public information | Human correction

HUMAN HANDOFF / PAUSE / EXIT

05

Accessible Slow-Mobility Task Chain

Barrier work-order closure | Paper and phone remain

HUMAN HANDOFF / PAUSE / EXIT

06

Multilingual Railway Culture Guide

Human historical review | No route tracking

HUMAN HANDOFF / PAUSE / EXIT

07

Public-Space Microclimate Assistant

Environmental advice | Site staff decide

HUMAN HANDOFF / PAUSE / EXIT

08

Low-Speed Robot First-Use Route

Bounded place, time and speed | Emergency stop

HUMAN HANDOFF / PAUSE / EXIT

09

AI-Native First Store

Product, after-sales and reuse | No profile pricing

HUMAN HANDOFF / PAUSE / EXIT

10

Innovation Procurement Clinic

Acceptance, cost and liability | No award promise

HUMAN HANDOFF / PAUSE / EXIT

11

First Mile Console

Gate, owner and pause state | Public view de-identified

HUMAN HANDOFF / PAUSE / EXIT

12

Reuse Stage and Failure Archive

Reuse conditions and failure lessons | Anonymous allowed

HUMAN HANDOFF / PAUSE / EXIT

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A scenario without a problem owner, human baseline, handoff owner or stop condition does not enter the field; scenario count never substitutes for real adoption.

90 Days: Prove One Real User Before Construction

Responsibility, output, cost, evidence release, quantity basis and stop condition enter one delivery ledger

90-DAY DELIVERY LEDGER | MONEY, DUTY, EVIDENCE

ORDER-OF-MAGNITUDE CONTROL, NOT A QUOTE | EACH GATE CAN PAUSE, RETURN OR EXIT



GATE	TIME	ACCOUNTABLE	MANDATORY OUTPUT	RELEASE	COST RMB 10K	ACCEPT / STOP
G0	D1–15	Problem owner	3 real tasks + human baseline	10%	15–25	Unobservable task / no exit → stop
G1	D16–35	Product + data owner	Rights, refusal, escalation, maintenance	25%	12–25	No data duty / safety fail → return
G2	D36–55	Site operator	Human roster + reversible kit + use record	25%	43–80	Ordinary route blocked / no handoff → stop
G3	D56–75	Independent operator	Takeover drill + unit cost + correction closure	25%	15–30	Takeover fail / complaint open → return
G4	D76–90	Reuser + buyer	Acceptance, fee, liability, exit + reuse pack	15%	13–24*	No real reuse intent → archive; no build-out

DESIGN-CONTROL ENVELOPE | RMB 0.98–1.84M

- Research/co-design 15–25
- Human desk/training 18–30
- Reversible kit/furniture 25–50
- Access/wayfinding 15–30
- Testing/data security 12–25
- Contingency 15%: 13–24

THREE PROCUREMENT GUARDRAILS

1. No problem owner, no hardware
2. At least 2 quotes + site survey
3. Release follows evidence, not publicity

* G4 cost is the 15% contingency. Total is a concept control range; verify after official scope, quantities, procurement route and 2+ quotes.

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PILOT QUANTITY BASIS | WHERE RMB 0.98–1.84M COMES FROM

CONCEPT CONTROL, NOT A QUOTE | UPDATE QUANTITIES, UNIT RATES AND ROUTE AFTER SURVEY + 2 QUOTES



WP	CATEGORY	QUANTITY BASIS	UNIT / SUBITEM ASSUMPTION	LOW RMB10K	HIGH RMB10K
W1	Research / co-design	1 human baseline + 12 fixed interviews + design coordination	Baseline 6–10; interviews 3–5; coordination 6–10	15	25
W2	Human desk / training	80 two-person service shifts + 6 training / takeover drills	Shifts 12–19.2; training 6–10.8	18	30
W3	Reversible kit / furniture	36 m² demountable service kit + user table / sign kit	Kit 18–32.4; furniture/signs 7–17.6	25	50
W4	Access / wayfinding	1 route audit + reversible repairs + 12 field information signs	Audit 3–5; repairs 8–18; signs 4–7	15	30
W5	Testing / data security	1 rights clearance + 1 evaluation suite + 1 privacy/security review	Rights 3–6; evaluation 4–8; security 5–11	12	25
W6	15% contingency	Approved-scope uncertainty only	Rounded from W1–W5 subtotal	13	24

CONTROL TOTAL RMB 0.98–1.84M

PERMANENT WORKS, LAND, BULK DEVICES: OUTSIDE THIS PILOT

THREE DECISION PREREQUISITES

SITE SURVEY + QUANTITIES + 2+ QUOTES PER ITEM

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RMB 0.98-1.84M is concept control for one first pilot, not a quote or fiscal promise; update after survey, quantities, procurement route and 2+ quotes.

Long-Term Operation and a Global Public Brand Asset

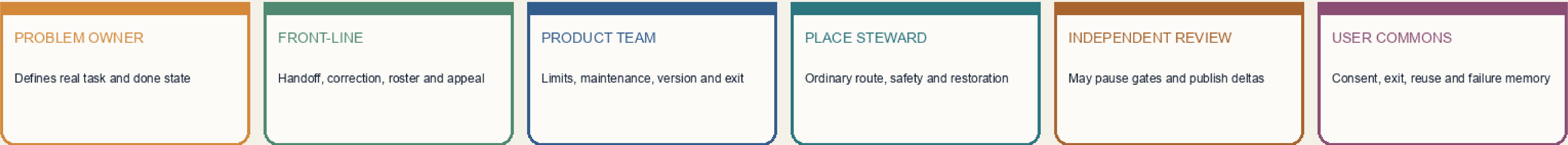
Six parties, dual ledgers, four seasons and three years; verified reuse matters more than city logos

LONG-TERM OPERATIONS | MAINTAIN AN ADOPTION LINE, NOT A ONE-OFF EVENT

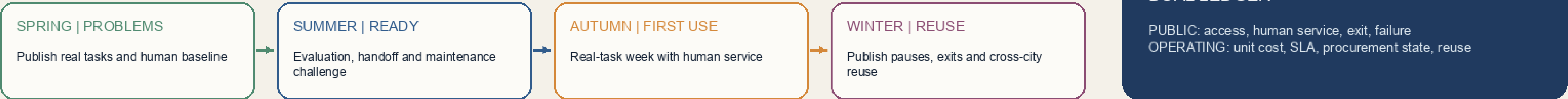


ROLES, ANNUAL RHYTHM, DUAL LEDGER AND THREE-YEAR ROADMAP; ORGANIZATIONS AND FUNDING REQUIRE FORMAL CO-CREATION

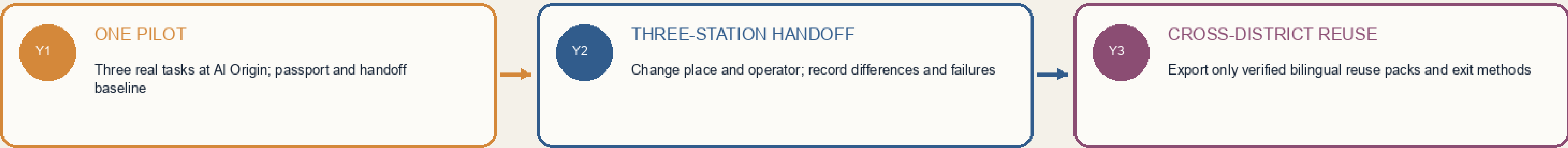
SIX-PARTY STEWARDSHIP: SEPARATE DELIVERY POWER AND STOP RIGHTS



ANNUAL RHYTHM



THREE-YEAR ROADMAP: SCALE ONLY AFTER REUSE EVIDENCE



STATUS BOUNDARY: OPERATORS, ANNUAL FUNDING, PARTNER CITIES AND EVENTS ARE CO-CREATION PROPOSALS, NOT COMMITMENTS.

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No party declares success alone; Year 1 proves operation, Year 2 replicates G4 services, and Year 3 connects cities through bilingual reuse packs and open interfaces.

FIRST-MILE PASSPORT | WHAT EARNS THE NEXT MILE

SYNTHETIC EMPTY EXAMPLE: REPRODUCIBLE STRUCTURE; ORG, PEOPLE, PERFORMANCE AND ORDER ARE NULL

FIRST-MILE PASSPORT
FM-DEMO-000 / SYNTHETIC_NON_OPERATIONAL
NOT STARTED / NO REAL RESULTS

- 01 PROBLEM OWNER**
NULL | NAME BEFORE G0
- 03 REAL TASK**
Find authority, human channel and complete one correction
- 05 DATA BOUNDARY**
Rights-cleared public info, no face or continuous trajectory
- 07 UNIT SERVICE COST**
NULL | CALCULATE AFTER PILOT
- 09 HUMAN DECISION DELTA**
PENDING | system advice / final decision / reason
- 02 FIRST USER**
NULL | NOT RECRUITED
- 04 HUMAN BASELINE**
PENDING | time, error, material, wait
- 06 HUMAN HANDOFF**
PENDING | visible desk + context export
- 08 EXIT & DELETION**
No-loss exit; retention set by accountable operator
- 10 REUSE CONDITION**
NULL | NO SUBSTITUTE WITH LIKES OR TRAFFIC

FIVE GATES ARE REVOCABLE STATES

- G0 REAL NEED
PENDING
- G1 PRODUCT READY
PENDING
- G2 FIRST USE
PENDING
- G3 FIRST ADOPTION
PENDING
- G4 FIRST REUSE
PENDING

IF A GATE EXPIRES, EVIDENCE FAILS OR APPEAL STAYS OPEN:
PAUSE / RETURN / EXIT

FIRST-MILE GATE: A BRAND FOR THE FIVE-GATE JOURNEY

MONO-COMPATIBLE, DISTANCE-LEGIBLE, READY FOR WAYFINDING, PASSPORTS, GATES AND EVENTS

FIRST-MILE GATE
JING-ZHANG FIRST MILE
MAKE REAL ADOPTION HAPPEN

MARK CONSTRUCTION

- OPEN GATE
OPEN COLLABORATION
- RAIL / PATH
HERITAGE TO ADOPTION
- FIVE SLEEPERS
G0-G4, GO OR PAUSE
- FORWARD POINT
EVIDENCE BEFORE PROGRESS

FOUR APPLICATIONS

- 01 WAYFINDING PYLON
- 02 EVIDENCE PASSPORT
- 03 G0-G4 GATE SIGN
- 04 LAUNCH BADGE

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